BENCHMARK LANDSCAPE: AFFORD-Bench vs. LIBERO-PRO & LIBERO-X

A Systematic Comparison of Physical Contact Affordances, Domain Diversity, Sensory Modalities, and Policy Universality

★ RSS 2027 SPOTLIGHT

LIBERO-PRO

Zhou et al., 2026 • Spatial & Visual OOD

Baseline

Task Breadth
130 Tasks across 4 Domestic Suites

Environments
Domestic Kitchen & Living Room scenes

Sensory Streams
Single RGB (128 × 128) low-resolution

Contact Physics
Nominal Only (Rigid grasp, fixed friction)

Action Prompts
Atomic single-step language instructions

Policy Target
VLA Models (OpenVLA, π0)

Scale / Engine
6,500 Demos (1.3M Steps) • Robosuite (~15 FPS)

Limited to Nominal Contact Physics • Domestic Tabletop Only

LIBERO-X

Wang et al., 2026 • Hierarchical L1–L5 Shifts

Prior SOTA

Task Breadth
600 Tasks across 5 Domestic Suites

Environments
Domestic Living, Kitchen, Dining spaces

Sensory Streams
Dual RGB (256 × 256) mid-resolution

Contact Physics
Nominal Only (Rigid grasp, fixed friction)

Action Prompts
Atomic language instructions with subtask tags

Policy Target
VLA Models (OpenVLA, Octo)

Scale / Engine
2,520 Demos (0.5M Steps) • Robosuite (~15 FPS)

No 3D Depth Sensing • Domestic Tabletop Only

AFFORD-Bench (Ours)

RSS 2027 • Physical Affordance Grounding & Policy Agnostic

★ REVOLUTION

Task Breadth
210 Defined Tasks across 7 Distinct Domains

Environments
Industrial, Commercial Culinary, Lab, Tabletop, Chaining

Sensory Streams
Dual RGB (640×660, 300×420) + 3D Metric Depth

Contact Physics
Full Dynamics: Friction μ ∈ [0.1, 1.2], Mass, Clearances

Action Prompts
Compound 2–3 Step Continuous Instructions (D6, D7)

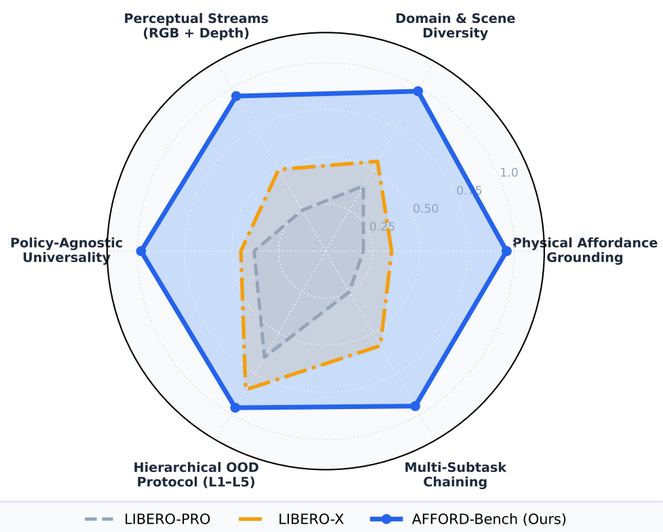
Policy Target
Policy-Agnostic: BC, DP, ACT, DP3, VLA, Offline RL

Scale / Engine
10,500 Demos (2.3M Steps) • C++ MuJoCo (45 FPS, $0 Cost)

✓ Full Contact Physics • Universal Policy Agnosticism • $0 Cost

(a) Multi-Dimensional Capability Radar

Normalized capability scores across 6 core robotic manipulation axes



(b) Quantitative Scale Comparison

Demonstration steps, evaluation suites, and domain diversity

1. Physical Demonstration Timesteps

+77% Scale

LIBERO-PRO	1.3M Steps
LIBERO-X	0.5M Steps
AFFORD-Bench (Ours)	**2.3M Steps (>10.5k Demos)**

2. Out-of-Distribution (OOD) Suites

+75% Suites

LIBERO-PRO	20 Suites
LIBERO-X	15 Suites
AFFORD-Bench (Ours)	**35 Suites (7×5 Regimes)**

3. Distinct Physical Domains

7 Real Domains

LIBERO-PRO	4 Domains (Domestic)
LIBERO-X	5 Domains (Domestic)
AFFORD-Bench (Ours)	**7 Diverse Domains**

(c) Core Technical Innovations

Four foundational breakthroughs introduced by AFFORD-Bench

#1 Physical Contact Dynamics & Slippage
Coulomb friction variations (μ ∈ [0.1, 1.2]), dynamic payload mass scaling (0.5–3.0×), and sub-cm clearances (±1.5–5 mm) expose rote visual trajectory memorization.

#2 High-Resolution 3D Depth Sensing
Dual-camera synchronized RGB (640×660 workcell + 300×420 wrist) paired with calibrated 3D metric depth point clouds, directly unlocking 3D spatial policies (DP3).

#3 Multi-Subtask Continuous Chaining
Domains D6 & D7 feature compound 2–3 step continuous instructions (e.g., 'align shaft, press bearing, lock clamp') to benchmark error compounding across extended horizons.

#4 Universal Policy Agnosticism
Standardized Robomimic HDF5 directly ingestible by BC, Diffusion Policy, ACT, DP3, OpenVLA, SmolVLA, AtomVLA, and Offline RL (CQL/IQL) with pre-computed splits.